QUESTIONNAIRE RESPONSE

Academic-Industrial Collaborations in Arab Countries

A Response to Dr. Frank Höselbarth

Nxtgn Academic-Industrial Cooperations Ltd.

Submitted by

Ahmed Mahmoud Saeed Ahmed Ali

Humanities Educator | Author | Member, British Philosophical Association

October 2026

Dear Dr. Höselbarth,

Thank you for this thoughtful questionnaire and for the opportunity to share my perspective on the academic and industrial landscape in the Arab world. The questions you raise touch on structural challenges that are rarely discussed with the candor they deserve. My responses are informed by my experience as an educator who has taught in Egypt and the United Arab Emirates, and who has observed the gap between educational theory and professional reality from both sides of the classroom. I offer these reflections not as definitive answers, but as an honest contribution to a conversation that I believe is long overdue.

I. Universities and Partnerships

1. In your view, what role do Arab universities play in building a knowledge-based society in the Middle East?

In my experience, the contribution of Arab universities to a genuinely knowledge-based society remains limited, and the reasons are structural rather than incidental. The most fundamental challenge is that the dominant epistemological framework in the region is, at present, imported. The knowledge we teach, the methodologies we apply, and the curricula we follow originate predominantly in Western academic traditions and are delivered in English or translated from it. Even after decades of globalization, this imported framework does not fully align with the cultural, mental, and religious fabric of Middle Eastern societies. The result is a persistent gap between what is taught in the lecture hall, even when supplemented with real-life examples, and what individuals encounter in their factories, schools, streets, and markets.

Until Arab universities bridge this gap, until they root their curricula in the identity and lived reality of the communities they serve, they will continue, in my view, to miss the point. A knowledge-based society is not built by importing knowledge alone; it is built by producing knowledge that speaks to the society it intends to serve. The role of Arab universities, therefore, should not be merely to transmit Western knowledge but to synthesize it with indigenous intellectual traditions and to produce graduates who can apply global frameworks to local realities with authenticity and competence.

2. How important do you consider strategic partnerships between industry and universities to be? Can you think of an example of a successful academic-industrial collaboration?

Strategic partnerships between industry and universities are, in principle, extremely important. They are the bridge between theoretical instruction and professional competence. However, the core issue we face in the Middle East is the systemic misuse of internships and field placements. In many factories, companies, and schools, students are consumed as labor rather than cultivated as learners. They are assigned heavy workloads that would otherwise require a paid professional, and the result is an exhausted student who is physically and mentally drained. Pressured by the volume of tasks required to maintain their placement, such a student forgets to learn in the process. The internship becomes exploitation dressed as opportunity.

This problem is compounded by the oversupply practices of certain universities, which prioritize enrollment numbers and graduation rates over genuine competence. The pressure to show high success rates, to attract enrollment for the following year, and to secure tuition revenue leads institutions to qualify the unqualified. This devalues the qualification itself and undermines fair employment for those who genuinely earned it. In my own experience as an educator, I have been directed to pass every student, including those who are chronically absent and those who submit blank examination papers.

That said, I do not wish to present a uniformly negative picture. There are examples of successful academic-industrial collaboration, particularly in Egypt, where they occur chiefly in two professions: medicine and engineering. In these fields, students genuinely engage in hospitals and on construction sites, practicing under the observation and mentorship of experienced professionals. These models work because the supervision is real, the learning is structured, and the student is treated as a learner, not as free labor.

3. Which Arab countries are the most advanced in STEM education? Are there any model universities for these disciplines?

My direct experience in STEM education is limited to the United Arab Emirates, and what I have observed there gives me cause for concern rather than optimism. The infrastructure is impressive, the investment is substantial, and the facilities are world-class. However, the academic integrity of the outcomes is, in my view, not commensurate with the investment. A significant proportion of students outsource their projects to paid contractors or do not complete them at all, knowing that they will receive passing marks regardless. In my estimation, the percentage of students who genuinely complete their own projects with creativity and intellectual ownership does not exceed 5% in the best-

case scenario across schools in the UAE. This is not a failure of the students alone; it is a systemic failure of accountability.

I am not sufficiently informed about STEM education in other Arab countries to offer a comparative assessment. However, based on what I know of Egyptian government universities, the academic rigor is considerably higher, as these institutions generally do not allow students to pass unless they genuinely demonstrate competence.

4. In your opinion, are universities fulfilling their obligation to adequately prepare students for the job market?

No. What universities provide is an excellent holistic overview of how work should be done in principle. The abstract information students acquire is valuable as a foundation. However, the reality of the workplace includes shortcuts, fixers, favoritism, nepotism, and, in some contexts, racism. I do not suggest that these phenomena are unique to the Arab world; they exist everywhere. But students should at least be made aware of their existence and be equipped with the resilience and perseverance needed to navigate them. What happens instead is that students graduate with an idealized image of professional life and are then shocked, and in many cases traumatized, by the unfairness of hiring processes, by mistakes made by managers that no one can spotlight without risking termination, and by the vast distance between what they were taught and what they encounter. Universities owe students not only knowledge but also preparation for the human reality of the workplace.

5. Would you agree with the statement that finding a suitable job is one of the main challenges for college graduates in Arab countries?

Yes, I agree. It is, unfortunately, a widespread reality that many graduates struggle to work in their chosen profession. One finds millions who graduate from faculties of commerce working as cashiers in convenience stores, qualified engineers working as ride-share drivers, and trained teachers, researchers, and lawyers working in call centers. The examples are numerous. In the current landscape, it is considered rare to work in one's profession and pursue one's vocation without the advantages of favoritism, nepotism, or the financial means to establish oneself independently. This represents not only a waste of individual potential but also a systemic failure to align educational output with economic need.

6. What improvements would be needed in degree programs and curricula to better prepare students for industry's requirements? Would it be helpful, for example, if industry experts supplemented the teaching?

There must be a mandatory period of on-field training, ideally one semester or one full year, integrated into every degree program and in every profession. This training must be supervised, and the feedback of the students must be actively sought and acted upon, so that they are not abused, overworked, or humiliated during the process. The involvement of industry experts in teaching, both during the on-field training and in the preparatory phase preceding it, would be an excellent addition.

Furthermore, curricula must be honest about the professional landscape they prepare students for. If a department of anthropology studies aboriginal peoples, there must be an allocated budget for students in that department to travel and meet those communities, rather than spending four years looking at photographs and watching videos. And if a country does not hire researchers or has no professional demand for them, it should seriously reconsider whether a faculty of arts should exist at all in its current form. To enroll young people in programs with no viable professional outcome is to waste their effort and their energy, which the country could otherwise direct toward its genuine developmental needs.

7. Is research and development sufficiently application-oriented?

Not at all. In my observation, many professors, if not the vast majority, pursue research and development primarily as a means to secure promotion, a pay raise, or a better academic title. The focus is rarely on the societal impact or the practical outcome of the research. This is a significant problem, because research that does not aspire to benefit society is, in a very real sense, research that has lost its purpose. The incentive structures in Arab universities, as in many universities worldwide, reward publication volume and citation metrics rather than genuine societal contribution. Until the incentive structure changes, the orientation of research will not.

8. How will AI change higher education?

It has already changed it. As a member of the British Philosophical Association, I correspond with professors in the United Kingdom, Spain, Egypt, the UAE, and Brazil, and the picture they describe is uniform and concerning. In most of these institutions, professors are now either willingly or under institutional pressure using artificial intelligence in their classes. But the problem does not stop there. Formative and summative assessments are being generated by AI. Projects and performance tasks are being evaluated by AI. The question is conceived by AI, the assignment is completed by AI, and the

evaluation is performed by AI. The entire cognitive loop, from design to execution to assessment, has been outsourced to the machine. The human element, which is the very substance of education, has been quietly removed from the process. This is not a future concern; it is the present reality.

9. In your view, what skills and experiences does industry value more today: even deeper IT knowledge combined with practical experience, or more soft skills such as critical thinking, communication skills, and the ability to work in a team?

In my observation, the era that valued soft skills as a distinct premium has passed. What industry values today is pragmatism at its peak. In education, the valued professional is the one who can teach the highest number of students without complaint while delivering strong results. In engineering, it is the one who can manage the greatest number of projects and deliver them as quickly as possible at the lowest budget. The skill that industry values most, therefore, is not a single competency but the capacity for cross-functional multitasking. Industry seeks individuals whose knowledge spans multiple domains, so that instead of hiring three or four specialists, it can hire one person who can execute all functions swiftly. This is not necessarily a positive development for the quality of work or for the well-being of the worker, but it is the current reality.

10. How open are universities to partnerships with companies? Do you know of an example of successful collaboration between universities and industry?

There are many examples of collaboration, but the distinction I would draw is between collaboration that genuinely hounds the experience of the students and prepares them, and collaboration that uses students as free labor. In the first model, which is rare, students are mentored, given structured tasks, and assessed on their learning. In the second model, which is prevalent, students are assigned the work that a paid employee would otherwise do, without proper guidance, without structured learning outcomes, and often without the ability to meaningfully participate. In some countries, such as the UAE, the internship is largely a formality. Students are placed in companies but are not permitted to touch equipment, teach in classrooms, or engage in substantive work. It is done as a protocol, to satisfy a requirement, not to develop a competence. In other contexts, the opposite occurs: students are given real work without any of the learning protections that would make that work educational. Both models fail the student.

11. Could you imagine that stronger partnerships with German universities and companies could offer a strategic advantage for Arab countries, especially now?

In principle, yes. German academic-industrial models, particularly the dual education system, are among the most effective in the world at bridging the gap between theory and practice. A genuine partnership that brought this model to Arab universities, with proper safeguards for student learning, could be transformative.

However, I must express a reservation that is not about Germany specifically but about the broader geopolitical context. The economic deterioration in many Arab countries, compounded by ongoing conflicts and the consequences of sustained external intervention in the Middle East, has produced a generation of young people whose primary aspiration is to leave. In such an environment, any partnership program that creates a pathway to study or work abroad will likely see a significant proportion of its participants choose not to return. The very instability that makes such partnerships necessary also makes their long-term sustainability uncertain. The youth who are most capable of benefiting from such programs are often the same youth who are most determined to emigrate. This is not a critique of the partnership itself but a recognition of the conditions under which it would operate. Any serious initiative must account for this reality and build incentives for return that are not merely economic but also social and intellectual.

II. Talent Pool and Tech Ecosystem

1. In your view, which Arab country has the best tech talent pool?

I would point to Egypt. My assessment is not based on national preference but on the academic culture of Egyptian government universities, which traditionally maintain rigorous standards and do not permit students to pass unless they genuinely demonstrate competence. In recent years, all faculties across Egyptian universities have implemented a requirement that no student may graduate without completing and passing a series of technology courses, with a pass threshold of approximately 70 to 80 percent. This creates a baseline of technological literacy that is, in my experience, more reliably enforced than in many other contexts in the region.

2. Where do you see the greatest need for development among the younger generation? In your opinion, which skills or experiences are most lacking?

Moral education. It has become increasingly rare to find students who demonstrate genuine respect for and appreciation of hierarchy, whether that hierarchy is based on age, knowledge, or professional standing. The younger generation possesses technical fluency and digital competence, but the foundational ethical formation that produces patience, humility, and respect for expertise is, in many cases, absent. This is not a failure of the students alone; it reflects a broader cultural shift in which speed and self-promotion are rewarded over depth and integrity. Any serious investment in the younger generation must address this dimension alongside the technical.

3. Do you believe there are enough suitable positions for high-potential IT professionals and, more generally, for STEM graduates in the industrial sector of Arab countries? If not, what would need to be done to create more jobs?

No, the market currently suffers from oversupply in many IT fields. However, the problem is not merely quantitative but structural. Many positions that exist are filled through networks of favoritism and personal connection rather than through merit-based competition, which means that talent is not the primary criterion for employment. To create more genuine opportunities, three things are needed: first, the diversification of Arab economies away from single-source dependency, whether on oil, remittances, or foreign aid, toward knowledge-intensive industries that generate indigenous demand for STEM talent. Second, the enforcement of transparent, merit-based hiring practices in both the public and private sectors. Third, the cultivation of a domestic startup ecosystem that is supported not by rhetoric but by genuine access to capital, legal protection, and market opportunity. Without these structural changes, the oversupply will persist, and the talent will continue to seek employment abroad.

4. How mature are the local AI and technology ecosystems?

When we speak of the Arabic world, the maturity of the AI and technology ecosystem must be measured by a single criterion: ownership. So long as Arab countries are not the founders, owners, developers, creators, and host of the AI and the technological infrastructure they consume, the ecosystem is not mature. So long as we depend on another country to supply its developers or to host its data infrastructure, we remain consumers, not producers. Maturity is not measured by the sophistication of the technology one uses but by the depth of the craft one possesses. A society that imports its technology is, in this respect, a society that has not yet matured.

5. How good are the IT infrastructures?

In the most advanced Arab countries, such as the UAE, the IT infrastructure is excellent on the surface. The speed is high, the coverage is broad, and the investment is visible. Yet it is, in a deeper sense, very poor, because the skill and the craft are not in Emirati hands. They operate the infrastructure, but they do not build it, maintain it at its core, or control its foundational architecture. In the event of a serious crisis, a war, or a situation in which every expatriate professional must return to their home country, the true state of the infrastructure would be revealed. What appears robust would, I suspect, prove fragile, because the knowledge that sustains it does not reside in the population that depends on it.

6. Are you aware of any measures taken by Arab governments to support the tech ecosystem and the young startup scene? Which ones?

There is no shortage of advertising. Arab governments frequently announce initiatives, programs, and visions for technological advancement. However, in my observation, these are often primarily mechanisms for the collection of taxes, fees, and regulatory charges, with no genuine structural support for the business from any meaningful dimension. A startup receives a logo, a press release, and a stall at a conference. It does not receive the one thing it actually needs: a genuine market with fair competition and protected intellectual property. The rhetoric of support is abundant; the substance of support is scarce.

7. In your view, is there already a culture of intellectual property in Arab countries?

Yes, indeed. There is a genuine culture of intellectual property in the sense that people understand and value the concept of ownership over ideas, brands, and innovations. The legal frameworks exist. The awareness exists. What is lacking is not the culture but the enforcement. Intellectual property is understood but not consistently protected, and without protection, the culture remains aspirational rather than operational.

8. Where do you think the tech ecosystem will be in five years?

I hold a view that may appear unconventional in this context. I believe that technology will continue to advance until it reaches a point at which it begins to undermine the very foundations of its own sustenance: the human cognitive and social capacities that produced it. At that point, a correction will occur, and what will emerge is a return to the essential human crafts. The competencies that will matter most will not be those of the programmer or the data scientist but those of the person

who can plant, build, communicate face to face, read the stars, and think deeply without a screen. Technology will not disappear, but it will cease to be the center of human aspiration. The individuals who retain the craft of being fully human, of engaging with the physical and the natural world with direct competence, will be the ones who hold genuine value. This is not a rejection of technology; it is a recognition that technology, when taken to its logical extreme, dissolves the very conditions that make it meaningful.

III. Business Environment and Risks

1. What challenges do foreign companies face when establishing themselves in Arab countries?

Foreign companies face several serious challenges. The first is the risk of fraudulent practices by certain government employees or intermediaries who exploit the foreign company's unfamiliarity with local bureaucratic procedures. The second is the language barrier: some foreign investors begin operations without adequate command of the local language, which creates dependencies on local agents who may not always act in the company's best interest. The third is the opacity of regulatory frameworks, which are often applied inconsistently and can change without notice. The fourth is the challenge of navigating patronage networks: in many Arab business environments, access to opportunity is mediated through personal relationships rather than through transparent processes, and a foreign company that does not invest in building these relationships may find itself excluded from contracts for which it is technically qualified.

2. What conditions must be met for success?

First, sincerity of intention. A foreign company that enters an Arab market solely to extract profit, without genuine commitment to the development of the local economy and workforce, will be identified and resisted. Second, sufficient capital with a dedicated crisis fund. Markets in the region can be volatile, and the company must be able to sustain operations through periods of uncertainty. Third, sustained focus: the company must resist the temptation to diversify prematurely and must build its core competence before expanding. Fourth, a contingency plan. The geopolitical and economic conditions in the region can shift rapidly, and any company operating without a clearly defined alternative strategy is exposing itself to existential risk. Finally, the cultivation of local talent. A foreign company that relies permanently on expatriate leadership will never achieve genuine integration or acceptance.

3. Are there risks that, in your opinion, foreign companies often underestimate when establishing a branch in an Arab country? Where in particular?

Yes. The most underestimated risk is the historical memory of colonization. We Arabs do not forget, and this memory is transmitted from generation to generation, both through the formal education system and through the informal oral tradition of family gatherings. The narrative of what was done to us by the colonizers is not abstract history; it is a living presence in the collective consciousness. So while we may welcome the nationals of countries that once colonized us and committed grave injustices, there remains, in the hearts of many, a barrier to dealing with full sincerity. This is not directed at all foreign nations equally but specifically at those whose colonial history in the region involved the most severe transgressions. In Morocco, Algeria, Tunisia, Syria, Lebanon Mauritania, and Egypt, this memory is primarily associated with France. In Libya, it is associated with Italy. In Yemen, Iraq, Jordan, Palestine, Egypt, Kuwait, Bahrain, Qatar, United Arab Emirates, and Oman, it is associated with the United Kingdom. A foreign company from one of these nations that enters an Arab market without acknowledging this historical dimension, without demonstrating that it comes as a partner rather than as a successor to a colonial legacy, will encounter a reserve that it may not be able to name but that will nonetheless shape every interaction.

4. What regulatory and economic conditions should be given special attention?

I would draw particular attention to the principles of Islamic economics. A foreign company that takes the time to understand the Islamic economic framework will find it, as I believe, extremely fair and upright. It is a system designed to hold all parties accountable and to prevent the forms of exploitation that characterize purely profit-driven models. It prohibits usury, mandates risk-sharing, and places ethical obligations on commercial transactions that secular legal frameworks do not. A company that aligns its practices with these principles, even if it is not itself Muslim, will earn a degree of trust and goodwill that no amount of marketing can purchase. Conversely, a company that ignores these principles and operates as if the cultural and religious context were irrelevant will find itself met with quiet but persistent resistance.

5. How do you assess the current geopolitical stability in Arab countries?

So long as there is sustained external intervention from Western powers and the continued presence of an imposed state in the heart of the Arab world, the region cannot be stable. Stability is not merely the absence of active conflict; it is the presence of conditions in which a society can develop its institutions, its economy, and its culture without the constant disruption of external agendas.

These conditions do not currently exist in the region as a whole, and they will not exist until the fundamental causes of instability are addressed with honesty rather than with strategic interest.

6. How do you assess the economic situation in the Middle East? How do you differentiate between groups of Arab countries from an economic perspective? Does the younger generation believe in a bright future for themselves?

The economic situation varies significantly across the region. The Gulf states possess substantial wealth but are dependent on a single resource and on expatriate labor. The North African states possess human capital and agricultural potential but struggle with unemployment and political instability. The Levant and Iraq are contending with the consequences of prolonged conflict. Egypt, the most populous Arab nation, possesses extraordinary human capital but faces severe economic pressure.

As for the younger generation, we do not frame our outlook in terms of a bright or dark future. This reflects our faith. We believe that everything is written by Allah, and that fate is a music we shall dance to. Whatever is written, we will face it with a smile and with effort. We will hustle in life until we die. This is not passivity; it is a form of resilience that is deeply rooted in the Islamic worldview. We do not control what comes, but we control how we meet it. The younger generation may not believe in a bright future in the economic sense, but it believes in the sufficiency of its faith and in the dignity of its effort.

7. Which location in an Arab country would you recommend for a large technology center that creates jobs for college graduates?

If the objective is to find students who genuinely care about education by nature, who possess intellectual curiosity that is not manufactured by grades but rooted in character, I would highly recommend Egypt, northern Sudan, Algeria, Oman, and Jordan. These are societies in which the cultural value placed on education remains strong, in which the pursuit of knowledge is seen as a moral and not merely an economic good, and in which a technology center would find not only a willing workforce but a genuinely engaged one.

I hope these reflections are of use to your research. I am happy to elaborate on any of the points raised above, and I welcome the opportunity to continue this conversation.

With my sincere regards,

Ahmed Mahmoud Saeed Ahmed Ali

Humanities Educator | Author | Member, British Philosophical Association

ahmed@phronesis-studio.com | +63 9777 854 717